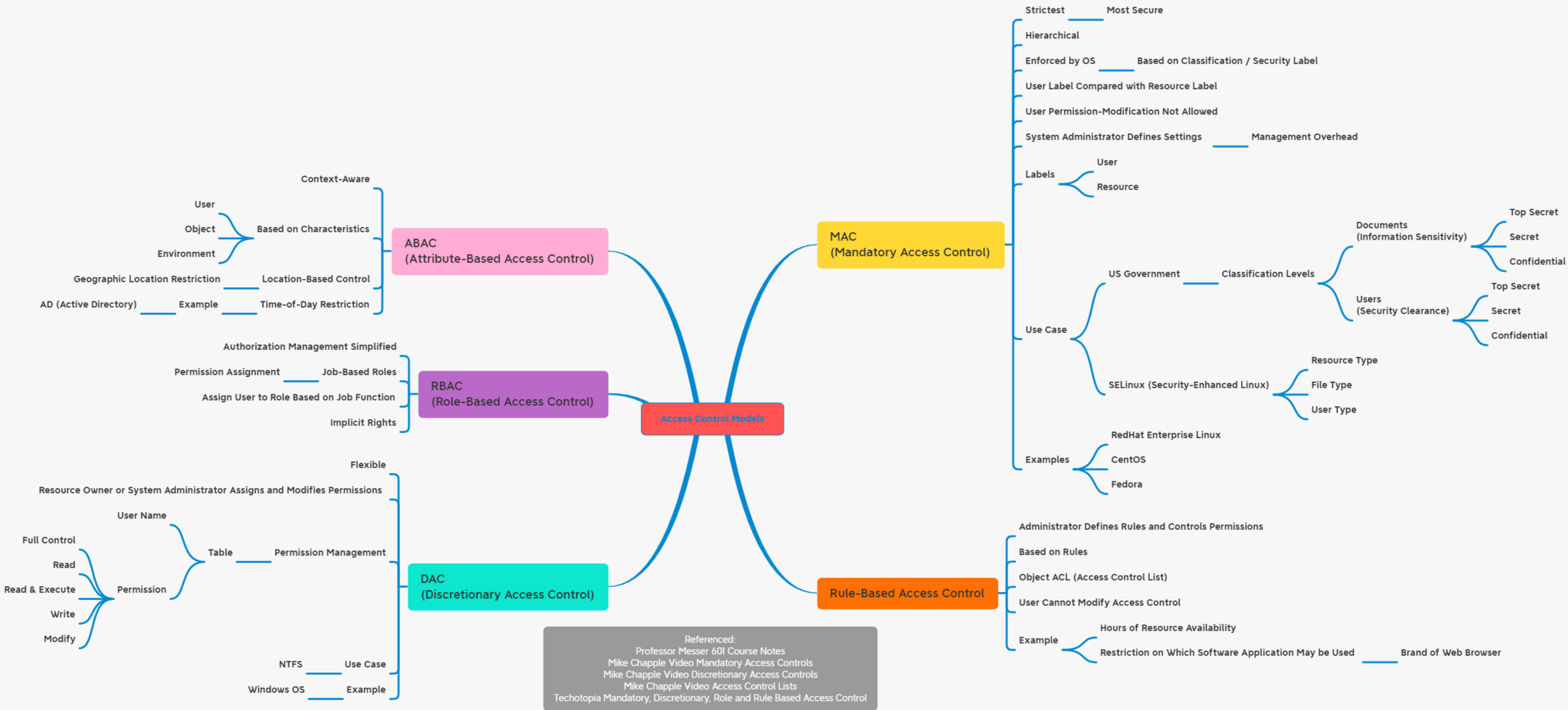
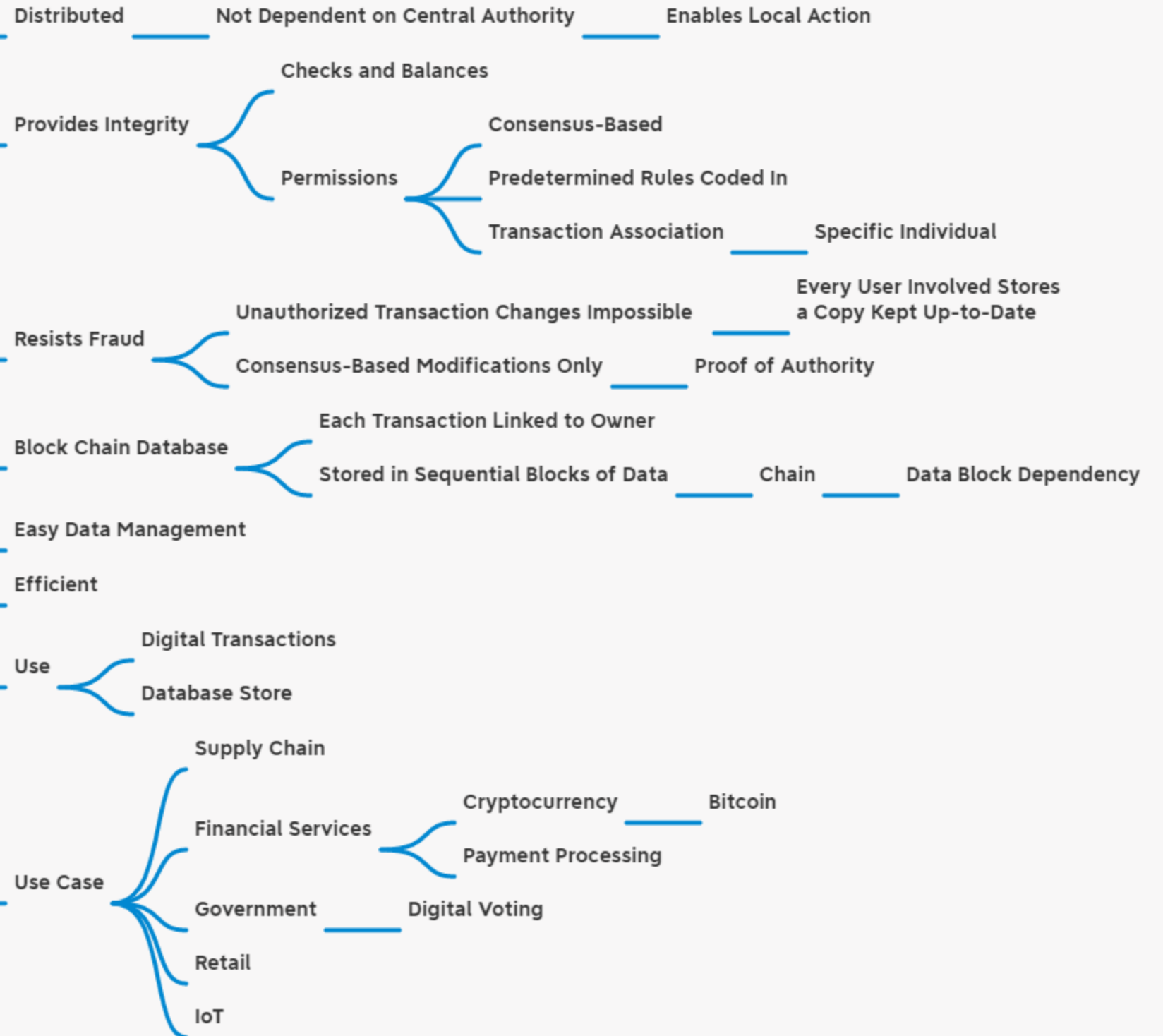


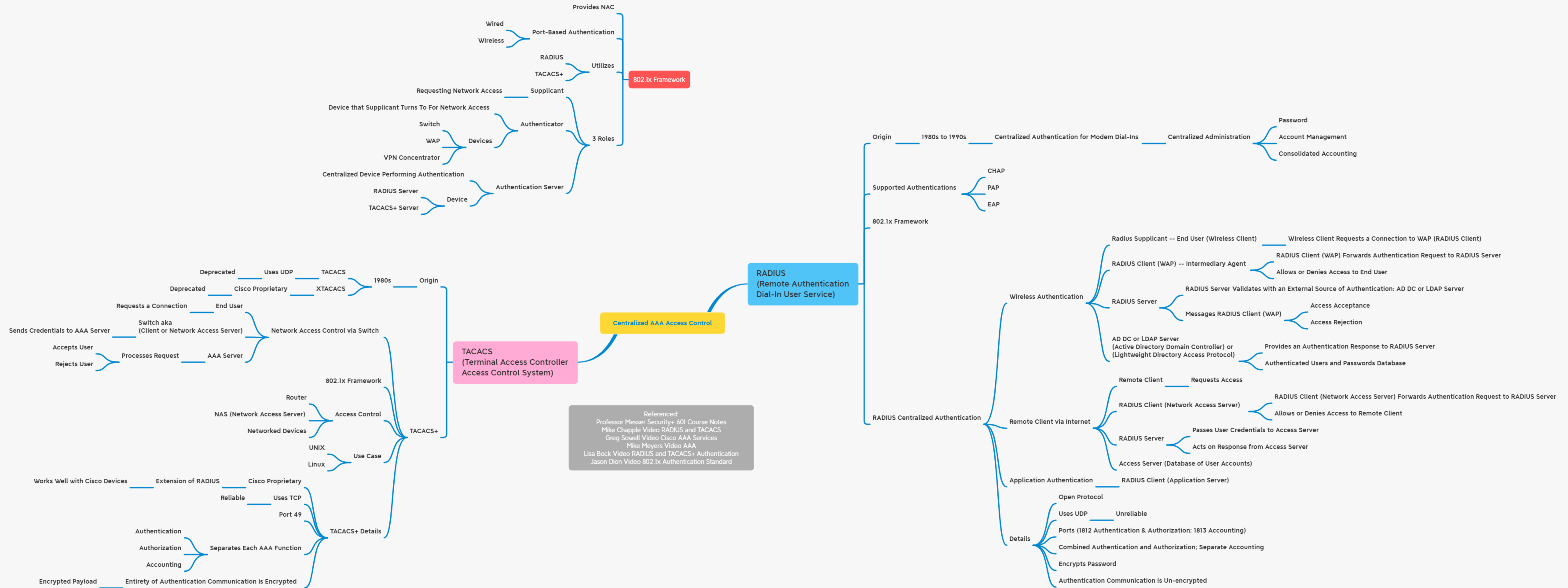


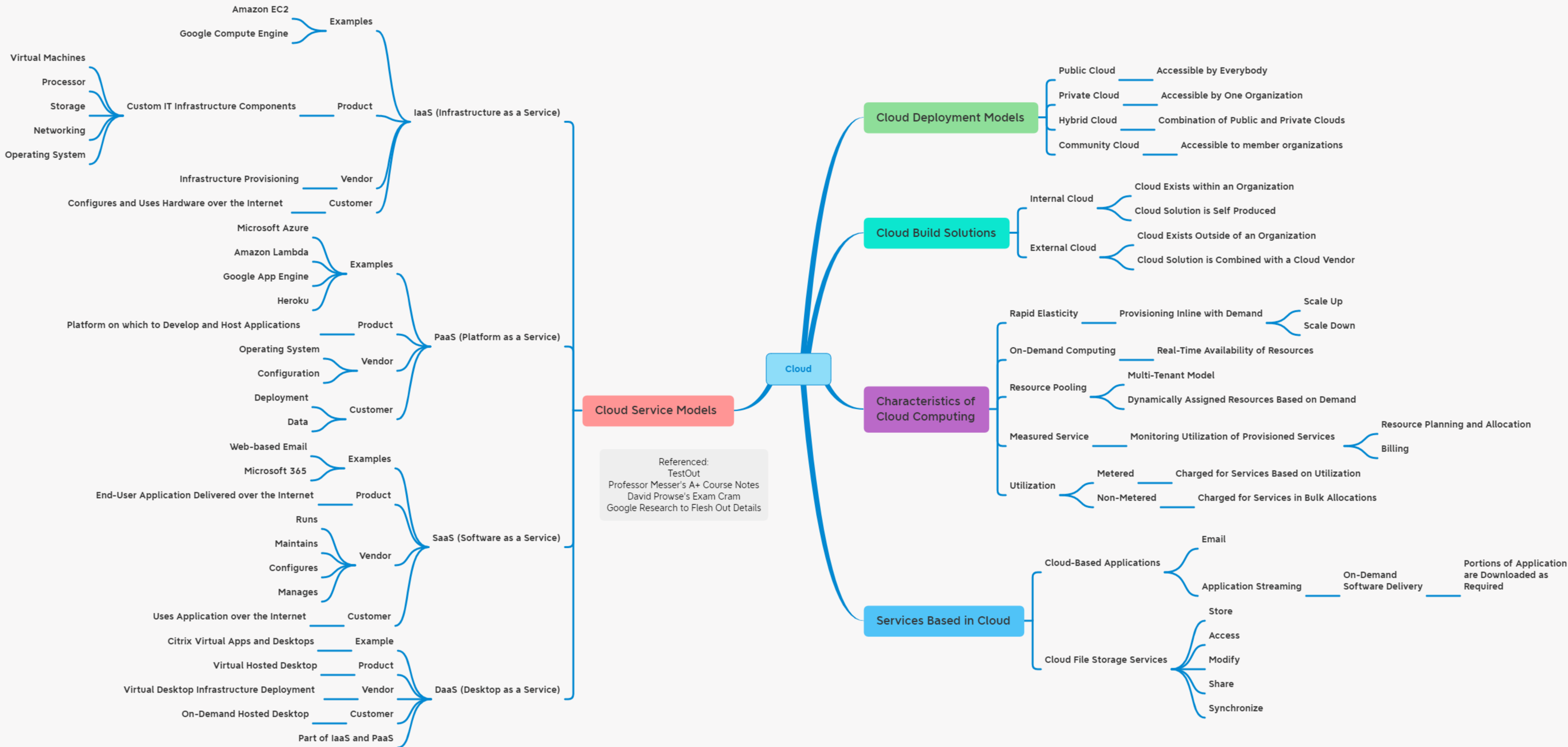
Blockchain
(In My Understanding)

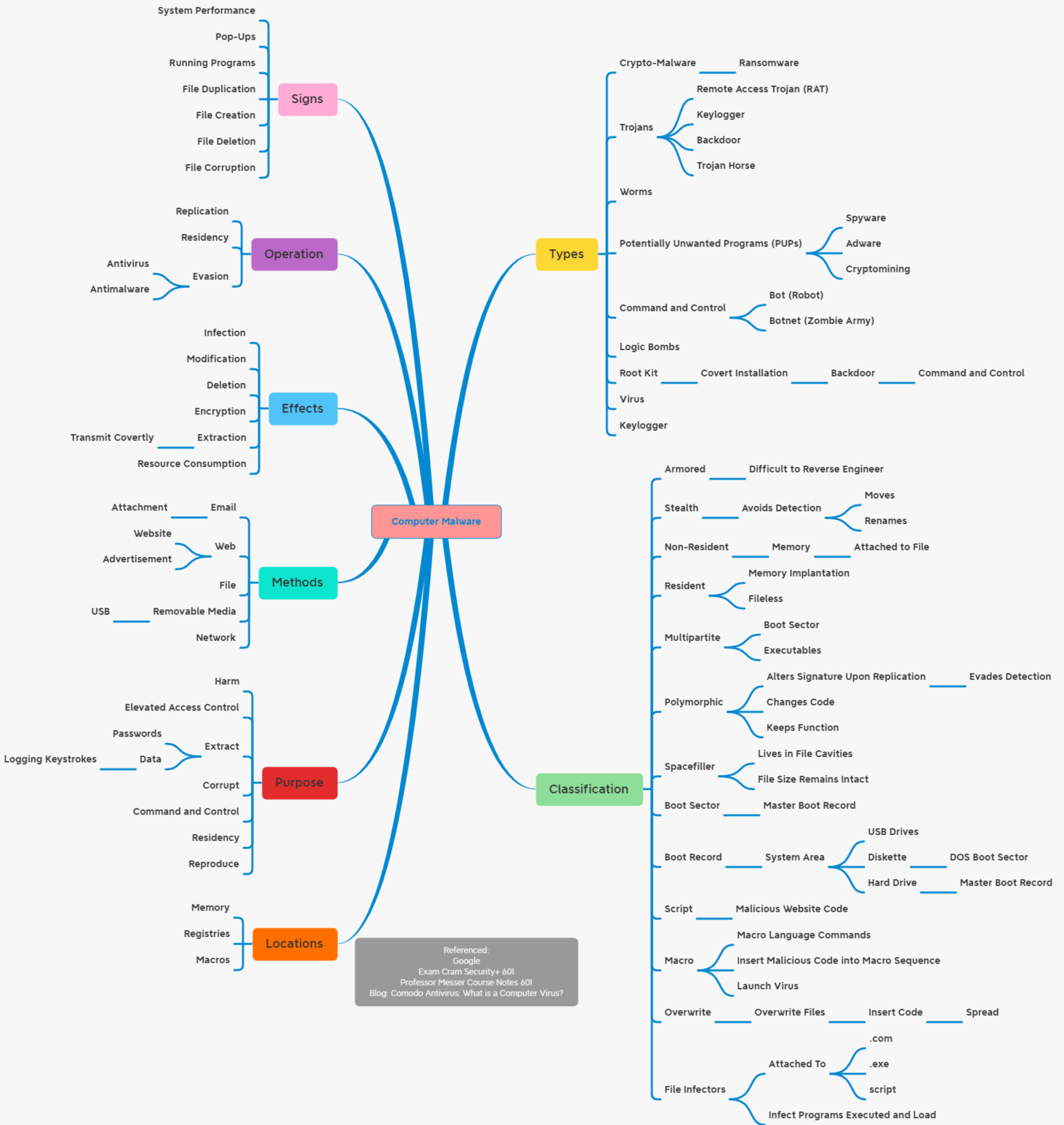
Distributed Ledger Technology (DLT)



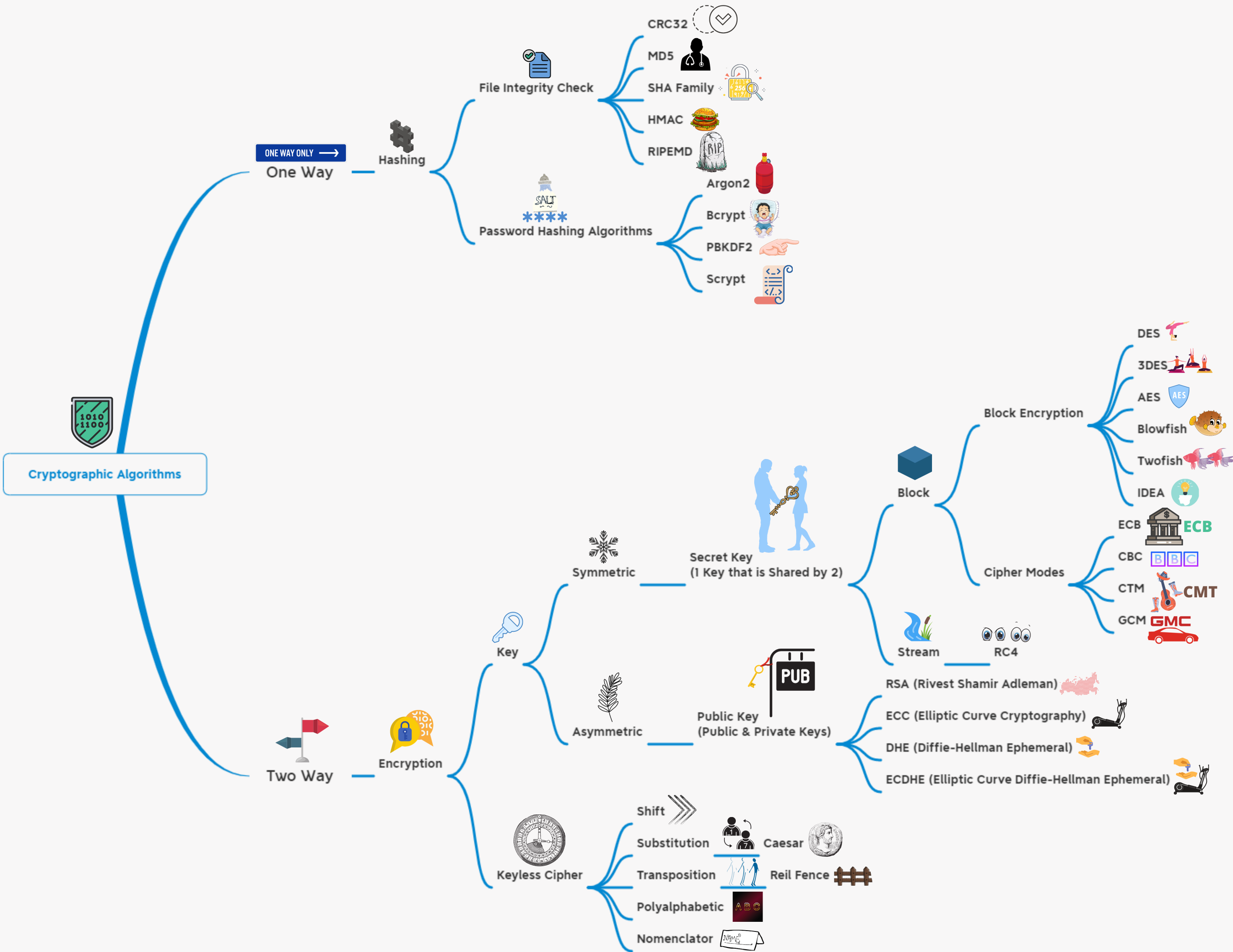
Referenced:
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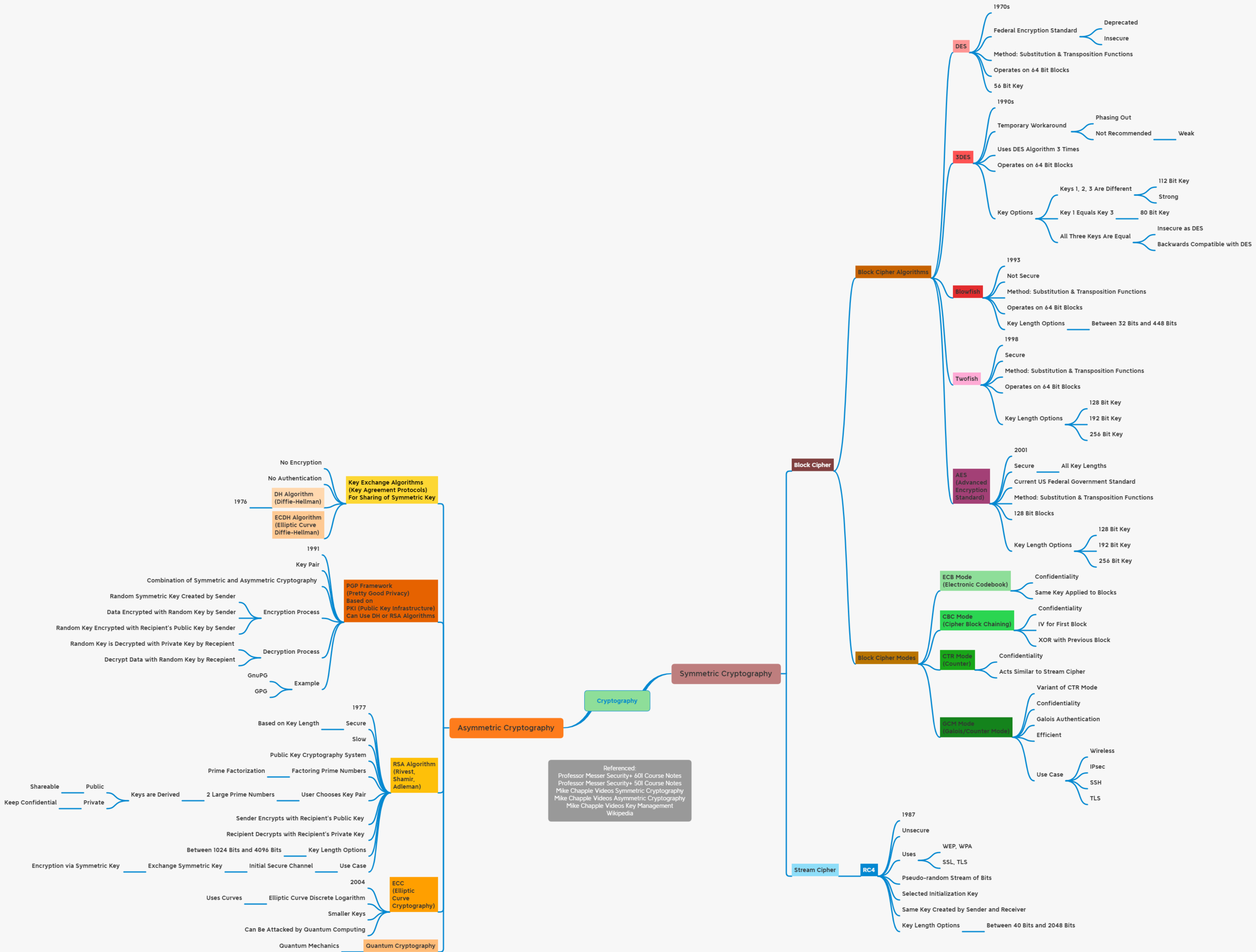


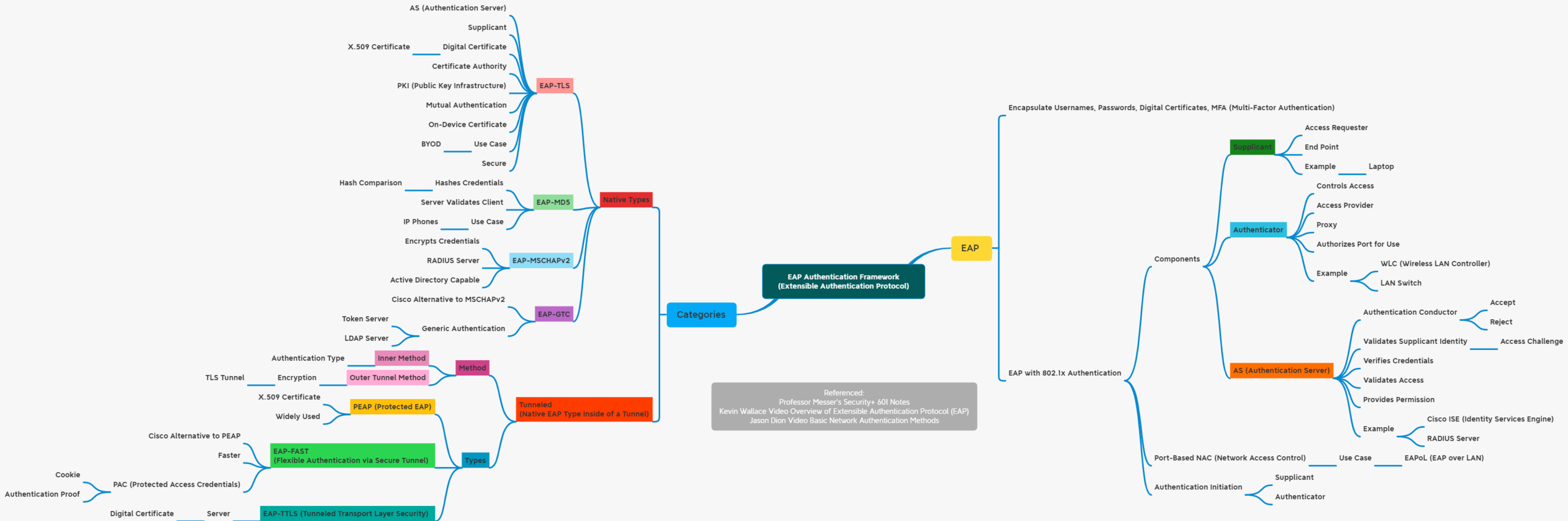


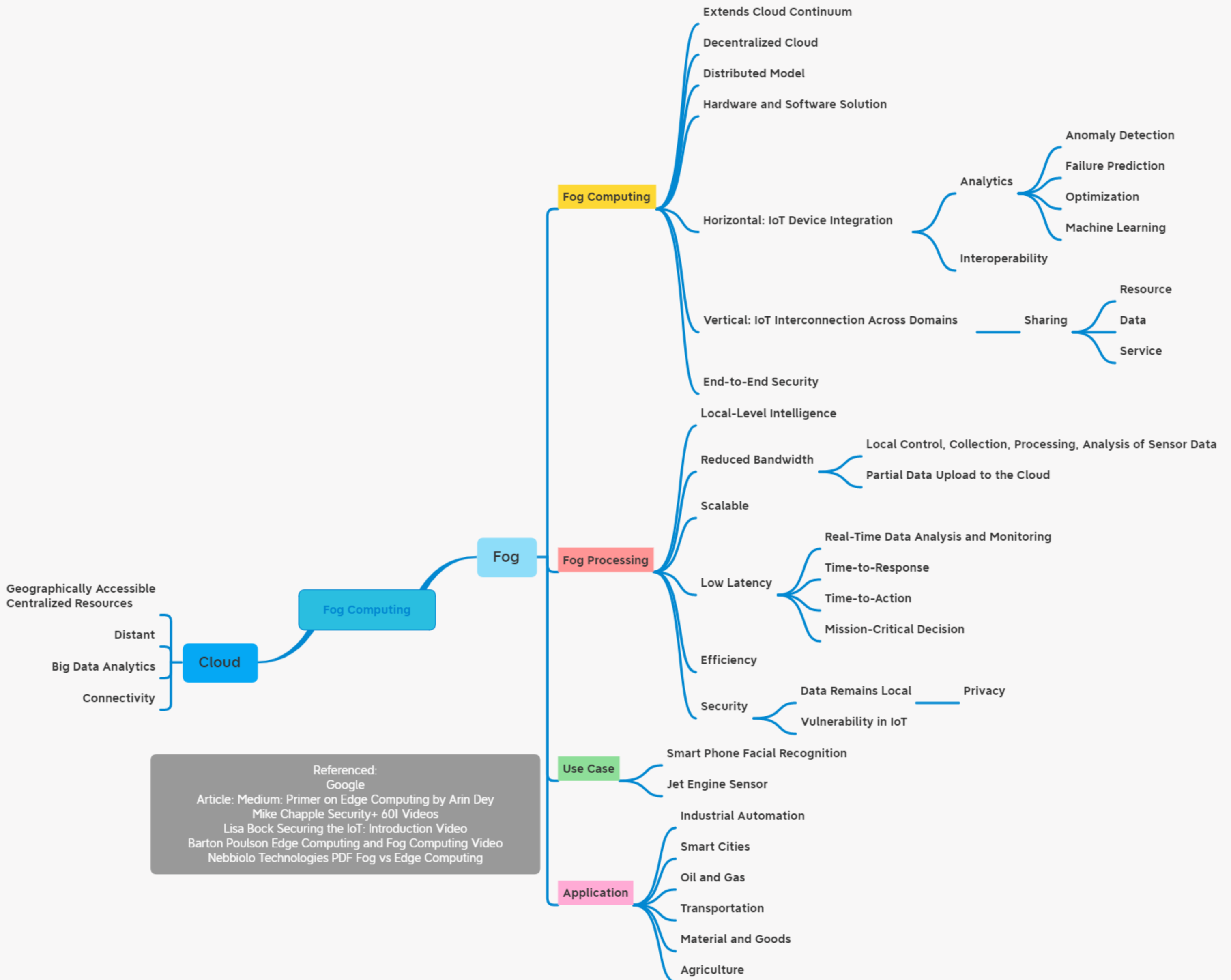


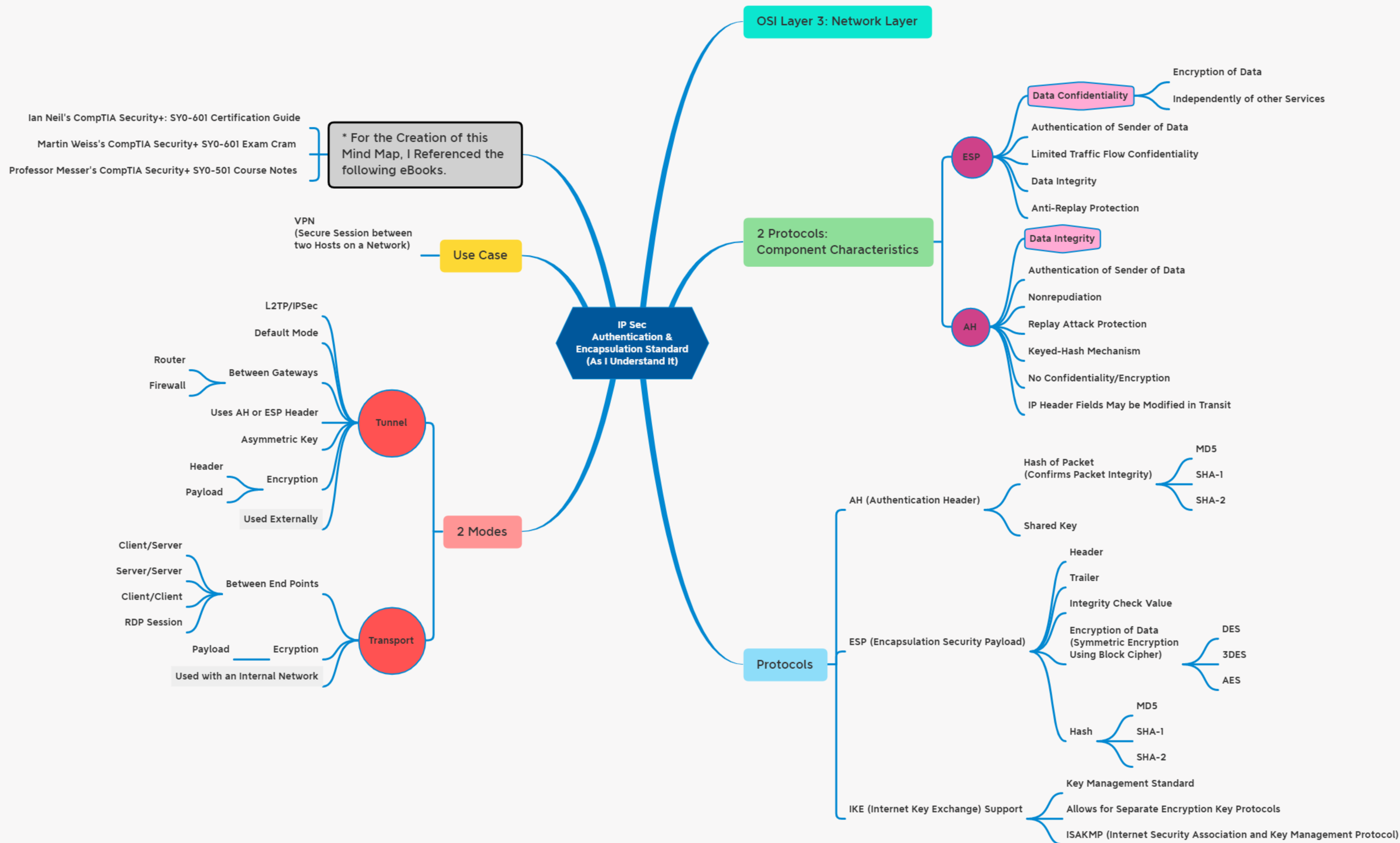
Referenced:
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Blog: Comodo Antivirus: What is a Computer Virus?

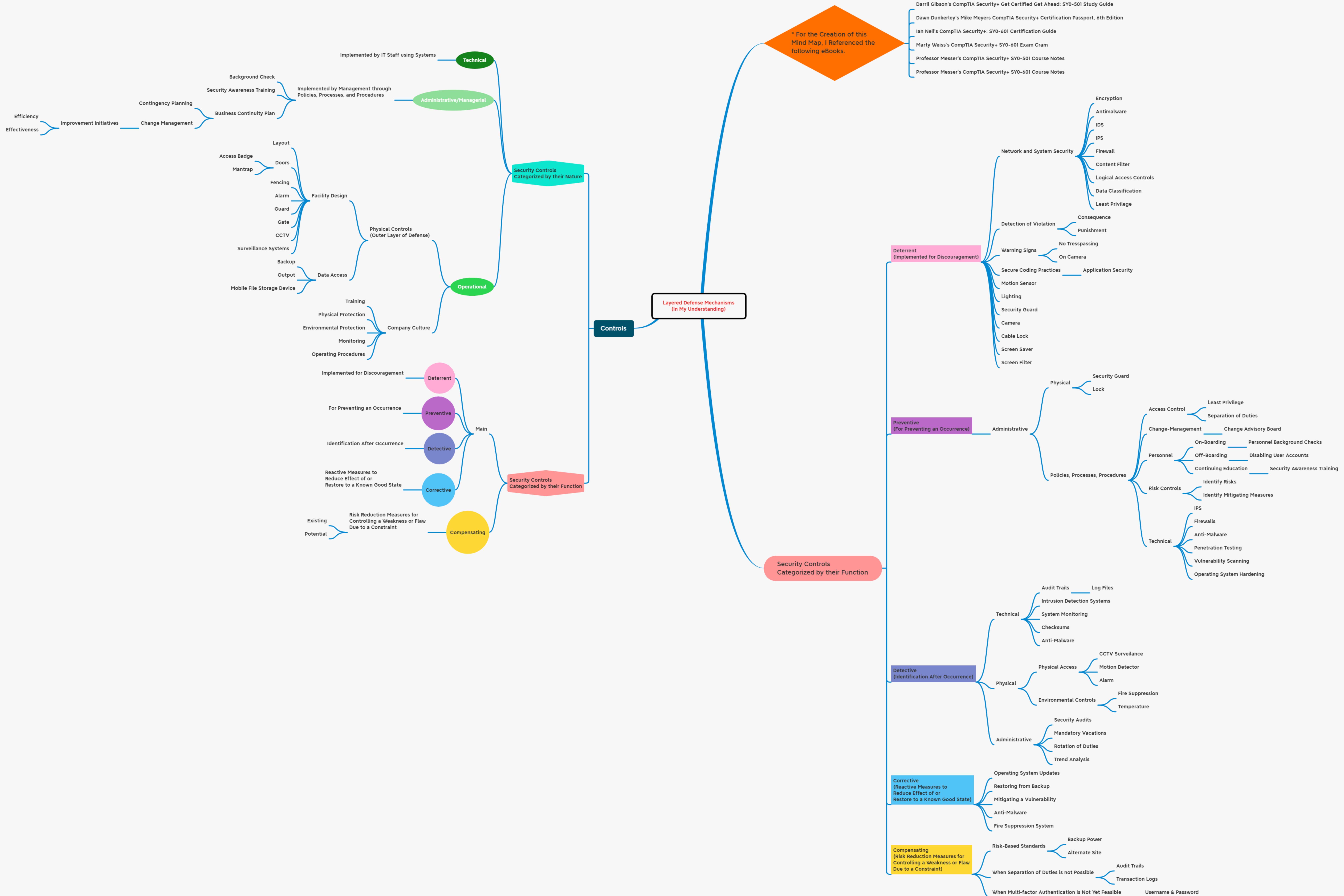


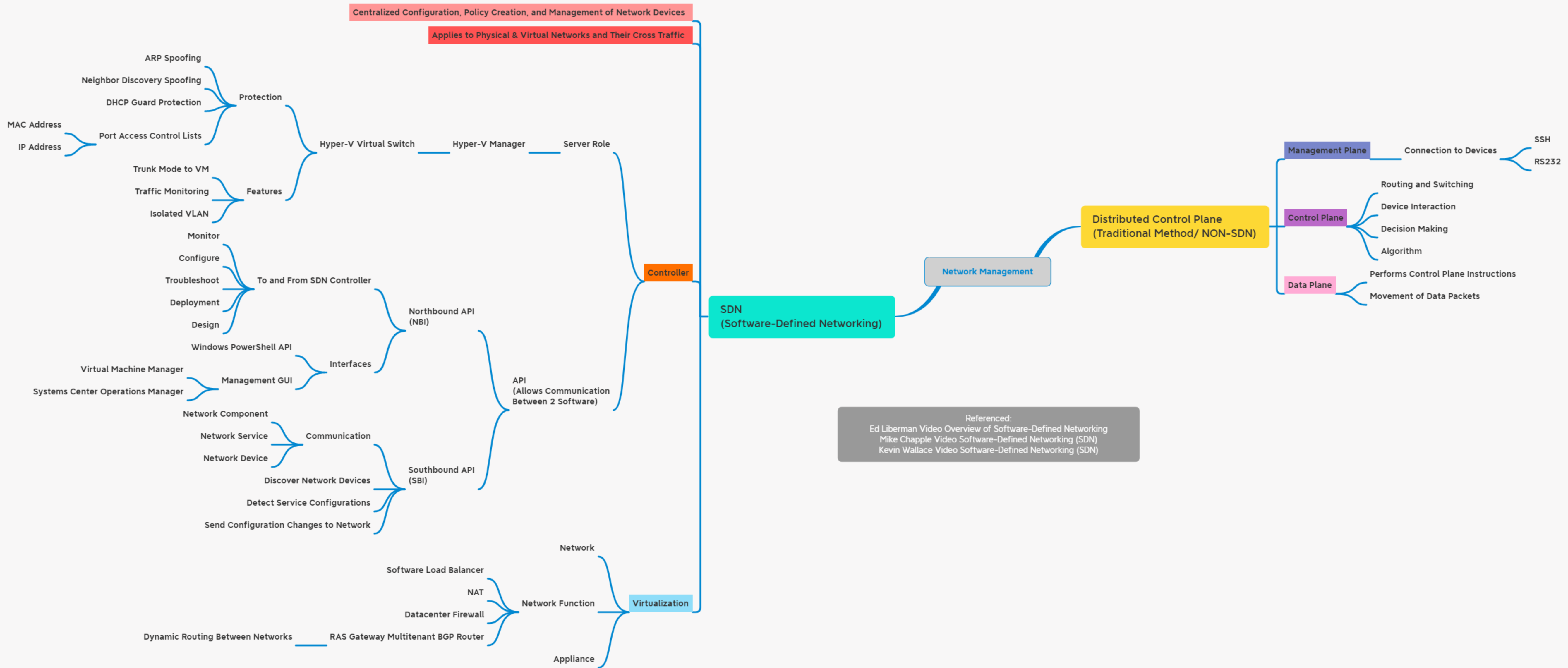


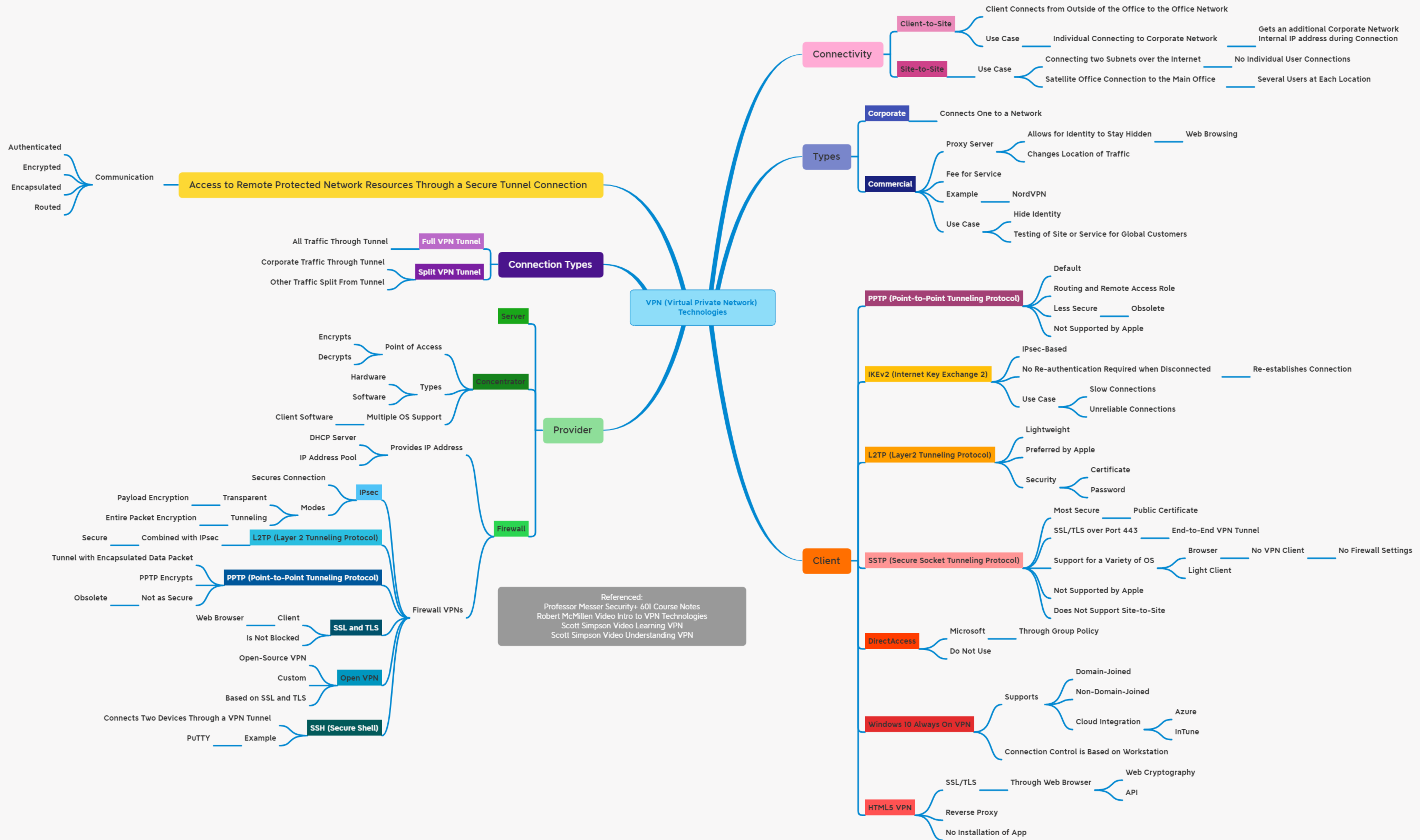




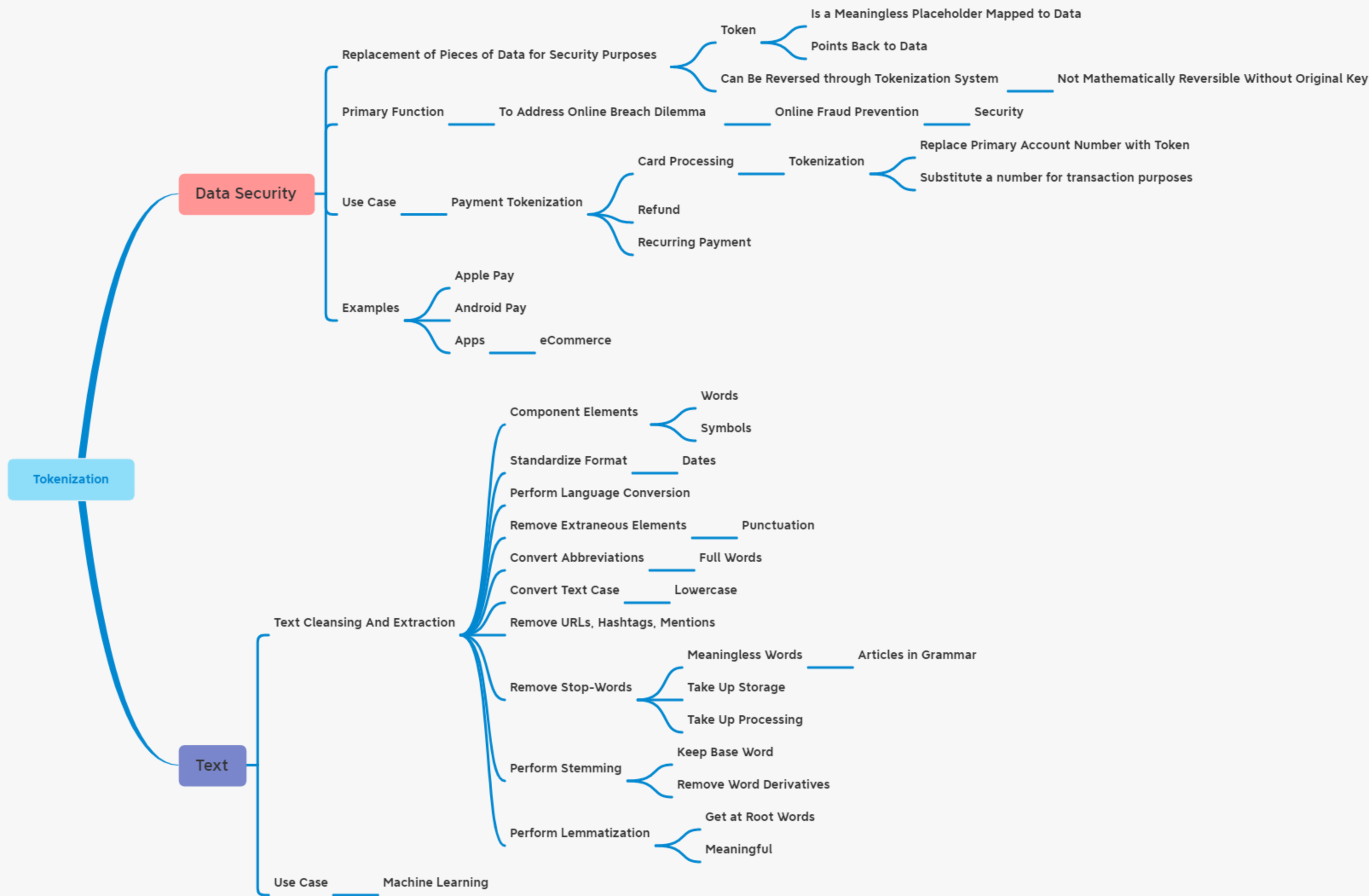






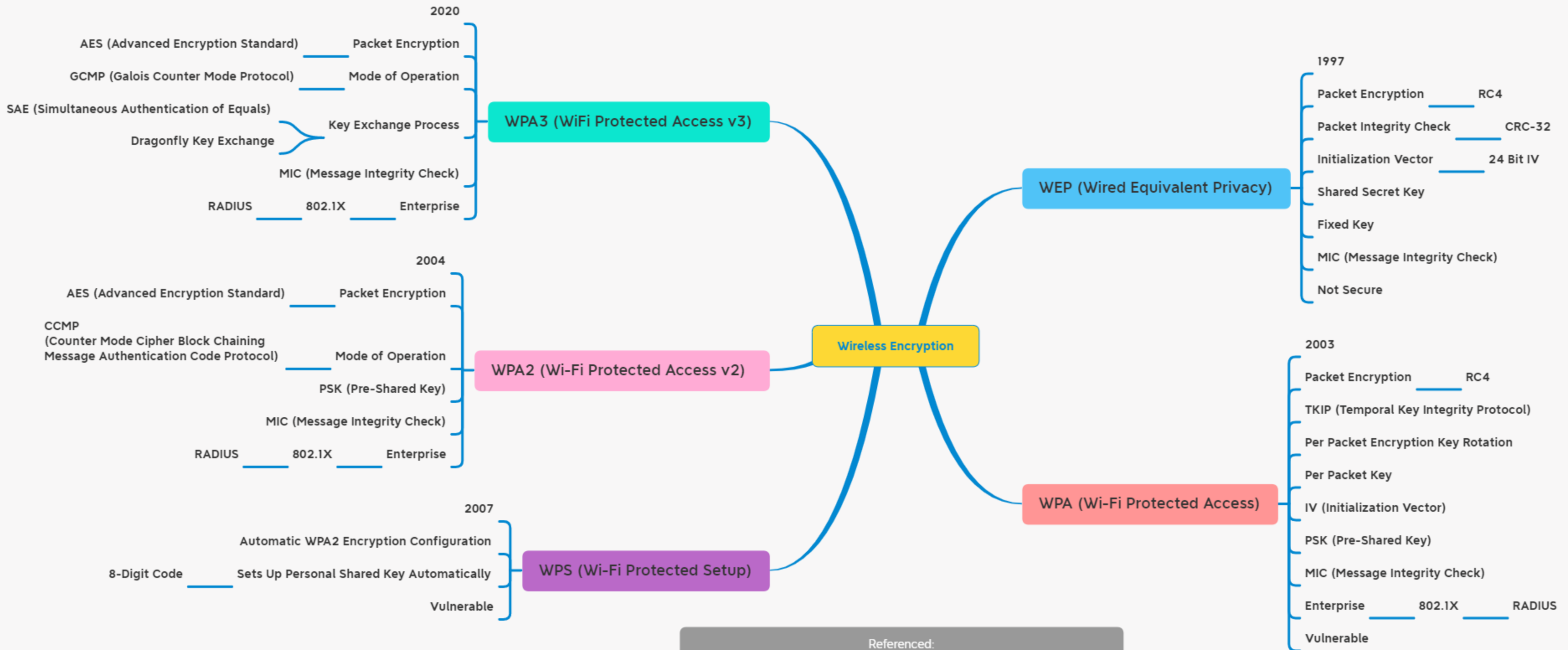


Referenced:
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Robert McMillen Video Intro to VPN Technologies
Scott Simpson Video Learning VPN
Scott Simpson Video Understanding VPN



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Mike Chapple Video Wireless Encryption
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Mike Chapple Video WEP, WPA, WPA2
Cybarrior Blog Wireless Security protocols: WEP, WPA, WPA2 and WPA3
Wikipedia Wi-Fi Protected Access

Cryptographic Algorithms

Symmetric Block Encryption

	Key Size (Bit)	Methodology	Advantage	Disadvantage	Use Case
DES	56	64-Bit Block			Do Not Use
					Financial Transactions
3DES	56, 112, 168	DES Algorithm		Slow	Alternative to AES
		64-Bit Block		Multiple Pass Encryption/Decryption	
		3 Phases			
		Multiple Keys		Resource Intensive	
AES	128, 192, 256	128-Bit Block	Fast		Small Devices
			Efficient		USB Drive Encryption
		One Pass Encryption/Decryption	Less Resource Intensive		CCMP WPA2
			Strong		TLS
Blowfish	Between 32 and 448	64-Bit Block	Faster than AES-256		Replacement for DES
Twofish	Between 40 and 2048	128-Bit Block			
IDEA	128	64-Bit Block			Replacement for DES
					Financial
					Government
					Broadcasting
					Audio/Visual

Cryptographic Algorithms

Block Cipher Modes

	Methodology	Advantage	Disadvantage	Use Case
ECB	Divide into Blocks		Weakness	Deprecated
	Encrypt Blocks		Same Key Per Block	
CBC	Combines Block with Previous Block		Inefficient	Symmetric Block Cipher
	IV for Randomness		Pipeline Delays	
	XOR Operation			
CTM	Block into Stream Cipher	Multiprocessing		Secure Mode of Operation
	IV & Counter			
	Different Key Per Block			
GCM	Counter Mode Operation	Efficient		Encryption / Decryption
		Performance		Authenticity
	Galois Mode Authentication			Integrity

Cryptographic Algorithms

Asymmetric Encryption

	Bit	Methodology	Advantage	Disadvantage	Use Case
RSA Rivest Shamir Adleman	2048	Prime Number Properties	Strong Security with Sufficient Key Size		Data Transmitted over Internet
		Based on Diffie-Hellman Static Key			Email
ECC Elliptic Curve Cryptography	Generates Key	Elliptical Curve	Less Resource Intensive		Low-Power Devices
					Small Wireless Devices
					Not Used by Gov Agencies
DHE Diffie-Hellman Ephemeral	Between 768 and 3072 Based on Group #s 1-15	Ephemeral Key		Negotiates for Strongest Supported Group	Sharing of Symmetric Key
	Higher Group # is More Secure				
ECDHE Elliptic Curve Diffie-Hellman Ephemeral	Generates Key Using ECC	Ephemeral Key		Negotiates for Strongest Supported Group	Sharing of Symmetric Key

Cryptographic Algorithms



File Integrity Check				
	Output Size Bit	Algorithm	Functionality	Use Case
CRC32		Checksum	Detect Data Corruption	
MD5	128	Message Digest	Verify Data Integrity	Email
				File on Disk
			Detect Data Corruption	File Download
				Executable File
SHA Family	0 (160)	Digital Signature	Encryption	Detection of File Modification by Malware
	1 (160)			
	2 (224, 256, 384, 512)		Data Integrity	HIDS
	3 (224, 256, 384, 512)		Authenticity	Antivirus
HMAC	HMAC-MD5 (128)	Shared Secret Key for Randomness	Data Integrity	IPsec
	HMAC-SHA1 (160)		Authenticity	TLS
RIPEMD		Message Digest	Verify Data Integrity	

Cryptographic Algorithms

Password Hashing Algorithms

	Bit	Salt	Functionality	Use Case
Argon2		Salt	Key Derivation Function	Backend Server
		Secret Key		Cryptocurrency
Bcrypt		Salt	Blowfish Encryption	Unix/Linux Shadow Password File
PBKDF2	128, 256, 512	Salt	HMAC Function	WPA2
				Apple iOS
				Cisco OS
Scrypt		Salt	Key Derivation Function	Wallets
				Files
				App Passwords

Wireless Standards

IEEE 802.11 Standard

	Frequency (GHz)	Streams (Up To)	Speed (Up To)	Channel Numbers (North America)	Channel Width (MHZ)	RF Modulation
802.11a	5	1	54 Mbps		20	CDMA, DSSS
802.11b	2.4	1	11 Mbps	1 through 11	20	CCK, DSSS
802.11g	2.4	1	54 Mbps	1 through 11	20	CCK, DSSS, OFDM
802.11n	2.4, 5	4 MIMO	150 x 4 = 600 Mbps	2.4: 1 through 11	20, 40	CCK, DSSS, OFDM
				5: 36, 40, 44, 48, 149, 153, 157, 161, 165		
802.11ac	5	8 MU-MIMO	866.7 x 8 = 6.933 Gbps	5: 36, 40, 44, 48, 149, 153, 157, 161, 165	20, 40, 80, 160	BPSK QPSK QAM